

MACSE622 – DATA ENGINEERING AND VISUALIZATION LAB

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Programme	MTech CSE (Big Data Analytics)	Slot	L33 + L34
Course	Data Engineering and Visualization Lab	Semester	2nd Semester
Date	05-01-2026	Exp No	3

Ex No: 3

Aim: Installation of Apache Kafka and performing Producer and Consumer Operations.

Version: 3.2.1 Scala 2.13

1. Started the Zookeeper batch file.

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.7462]
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C:\kafka>.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
[2026-01-05 10:44:34,416] INFO Reading configuration from: .\config\zookeeper.properties (org.apache.zookeeper.server.
quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,416] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,416] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,416] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,416] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (or
g.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DataDirCleanupManager)
[2026-01-05 10:44:34,433] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DataDirCleanupManager)
[2026-01-05 10:44:34,433] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DataDirCleanupManager)
[2026-01-05 10:44:34,433] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.z
ookeeper.server.quorum.QuorumPeerMain)
[2026-01-05 10:44:34,433] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2026-01-05 10:44:34,433] INFO Reading configuration from: .\config\zookeeper.properties (org.apache.zookeeper.server.
quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (or
g.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-01-05 10:44:34,433] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2026-01-05 10:44:34,433] INFO ServerMetrics initialized with provider org.apache.zookeeper.metrics.impl.DefaultMetricsP
rovider@2ea227af (org.apache.zookeeper.server.ServerMetrics)
[2026-01-05 10:44:34,433] INFO zookeeper.snapshot.trust.empty : false (org.apache.zookeeper.server.persistence.FileTxnSn
apLog)
[2026-01-05 10:44:34,448] INFO (org.apache.zookeeper.server.ZooKeeperServer)
```

2. Started the Kafka Server batch file.

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\kafka>.bin\windows\kafka-server-start.bat .\config\server.properties
[2026-01-05 10:46:56,702] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-01-05 10:46:56,861] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.X509Util)
[2026-01-05 10:46:56,909] INFO starting (kafka.server.KafkaServer)
[2026-01-05 10:46:56,909] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2026-01-05 10:46:56,925] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zookeeper.ZooKeeperClient)
[2026-01-05 10:46:56,925] INFO Client environment:zookeeper.version=3.6.3--6401e4ad2087061bc6b9f80dec2d69f2e3c8660a, built on 04/08/2021 16:35 GMT (org.apache.zookeeper.ZooKeeper)
[2026-01-05 10:46:56,925] INFO Client environment:host.name=AB3412-14.VITUCC.AC (org.apache.zookeeper.ZooKeeper)
[2026-01-05 10:46:56,925] INFO Client environment:java.version=21.0.1 (org.apache.zookeeper.ZooKeeper)
[2026-01-05 10:46:56,925] INFO Client environment:java.vendor=Oracle Corporation (org.apache.zookeeper.ZooKeeper)
[2026-01-05 10:46:56,925] INFO Client environment:java.home=C:\Program Files\Java\jdk-21 (org.apache.zookeeper.ZooKeeper)
[2026-01-05 10:46:56,925] INFO Client environment:java.class.path=C:\\kafka\\libs\\activation-1.1.1.jar;C:\\kafka\\libs\\aopalliance-repackaged-2.6.1.jar;C:\\kafka\\libs\\argparse4j-0.7.0.jar;C:\\kafka\\libs\\audience-annotations-0.5.0.jar;C:\\kafka\\libs\\commons-cli-1.4.jar;C:\\kafka\\libs\\commons-lang3-3.8.1.jar;C:\\kafka\\libs\\connect-api-3.2.1.jar;C:\\kafka\\libs\\connect-basic-auth-extension-3.2.1.jar;C:\\kafka\\libs\\connect-file-3.2.1.jar;C:\\kafka\\libs\\connect-js-on-3.2.1.jar;C:\\kafka\\libs\\connect-mirror-3.2.1.jar;C:\\kafka\\libs\\connect-mirror-client-3.2.1.jar;C:\\kafka\\libs\\connect-runtime-3.2.1.jar;C:\\kafka\\libs\\connect-transforms-3.2.1.jar;C:\\kafka\\libs\\hk2-api-2.6.1.jar;C:\\kafka\\libs\\hk2-locator-2.6.1.jar;C:\\kafka\\libs\\hk2-utils-2.6.1.jar;C:\\kafka\\libs\\jackson-annotations-2.12.6.jar;C:\\kafka\\libs\\jackson-core-2.12.6.jar;C:\\kafka\\libs\\jackson-databind-2.12.6.1.jar;C:\\kafka\\libs\\jackson-dataformat-csv-2.12.6.jar;C:\\kafka\\libs\\jackson-datatype-jdk8-2.12.6.jar;C:\\kafka\\libs\\jackson-jaxrs-base-2.12.6.jar;C:\\kafka\\libs\\jackson-jaxrs-json-provider-2.12.6.jar;C:\\kafka\\libs\\jackson-module-jaxb-annotations-2.12.6.jar;C:\\kafka\\libs\\jackson-module-scala_2.13-2.12.6.jar;C:\\kafka\\libs\\jakarta.activation-api-1.2.1.jar;C:\\kafka\\libs\\jakarta.annotat
```

Task 1: Performing the Producer Consumer operation:

3. Created a Kafka topic called test on port 9092, as producer
Send data from producer

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>kafka-topic.bat --create --bootstrap-server localhost:9092
'kafka-topic.bat' is not recognized as an internal or external command,
operable program or batch file.

C:\kafka\bin\windows>kafka-topics.bat --create --bootstrap-server localhost:9092 --topic test
Created topic test.

C:\kafka\bin\windows>kafka-console-producer.bat --broker-list localhost:9092 --topic test
>{"Name" : "John", "Age" : "31", "Gender" : "Male"}
>{"Name" : "Emma", "Age" : "27", "Gender" : "Female"}
>{"Name" : "Ronald", "Age" : "17", "Gender" : "Male"}
^
```

4. Created a Kafka consumer on the same localhost to display the ingested data.
Data is displayed at the consumer end

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>kafka-console-consumer.bat --topic test --bootstrap-server localhost:9092 --from-beginning
{"Name" : "John", "Age" : "31", "Gender" : "Male"}
{"Name" : "Emma", "Age" : "27", "Gender" : "Female"}
{"Name" : "Ronald", "Age" : "17", "Gender" : "Male"}
|
```

Task 2: Copying the contents of a file named “my_file.txt” from producer to consumer

5. Create a text file

```
my_file.txt X +
File Edit View H1 v ≡ v B I ↺ A
Hello World!
This is Bhaavya Ramita
I am a student at VIT Chennai
```

6. Exit the producer prompt and give the text file path along with the pipe operator followed by the producer command

```
C:\kafka\bin\windows>type "C:\Users\Student.412-14\Desktop\my_file.txt" | C:\kafka\bin\windows\kafka-c
onsole-producer.bat --broker-list localhost:9092 --topic test

C:\kafka\bin\windows>
```

7. The contents of the text file sent through the producer shall be displayed in the consumer end as follows

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>kafka-console-consumer.bat --topic test --bootstrap-server localhost:9092 --from-b
{"Name" : "John", "Age" : "31", "Gender" : "Male"}
{"Name" : "Emma", "Age" : "27", "Gender" : "Female"}
{"Name" : "Ronald", "Age" : "17", "Gender" : "Male"}
{"Name" : "Ronald", "Age" : "17", "Gender" : "Male"}
type my_file.txt | C:\kafka\bin\windows\kafka-console-producer.bat --broker-list localhost:9092 --topic
Hello World!
This is Bhaavya Ramita
I am a student at VIT Chennai
|
```

Show topics created:

8. To list all topics available in Kafka broker

```
C:\kafka\bin\windows>kafka-topics.bat --list --bootstrap-server localhost:9092
__consumer_offsets
test

C:\kafka\bin\windows>|
```

Streaming data with and without from-beginning

9. Creating a new topic “chat” and sending messages through producer

```
C:\kafka\bin\windows>C:\kafka\bin\windows\kafka-topics.bat --create --topic chat --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic chat.

C:\kafka\bin\windows>kafka-console-producer.bat --broker-list localhost:9092 --topic chat
>Hello, this is the first message
Second message coming through
```

Consumer end with from beginning

```
C:\kafka\bin\windows>kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic chat --from-beginning
Hello, this is the first message
Second message coming through
^CTerminate batch job (Y/N)? Processed a total of 2 messages
Y
```

Typing new message from producer end

```
C:\kafka\bin\windows>C:\kafka\bin\windows\kafka-topics.bat --create --topic chat --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic chat.

C:\kafka\bin\windows>kafka-console-producer.bat --broker-list localhost:9092 --topic chat
>Hello, this is the first message
Second message coming through
>>Third message now
>Fourth message after consumer started
>|
```

Consumer end without from beginning:

```
C:\kafka\bin\windows>kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic chat
Third message now
Fourth message after consumer started
|
```

Internal flow of Kafka Operations

1. **Zookeeper:** It acts as a coordination service, keeping track of kafka servers and their status. It keeps track of broker metadata.
2. **Kafka Server:** main Kafka process stored and serves messages. When kafka-server-start.bat is started, we launch a broker that listens to port 9092. It is now ready to accept producers/consumers.
They handle:
 - a. Receiving data from producers
 - b. Storing data in topics (organised by partitions)
 - c. Serving data to consumers
3. **Topic Creation:** A topic is where the messages are published. Brokers set up partitions or logs for it.
 - a. Internally, the topic is split into partitions (each partition is an ordered log)
 - b. Each partition is replicated across brokers for fault tolerance
 - c. When "test" is created, Kafka reserves space in its log files for that topic.
4. **Producer:** The producer sends messages to the broker.
 - a. Internally, the producer connects to the broker via TCP
 - b. It serializes the messages, the broker appends the message to the partition log of the topic. Each message gets an offset (a unique id within the partition)
5. **Consumer:** The consumer subscribes to the topic and reads messages.
 - a. Internally, the consumer connects to the broker. It requests messages starting from a certain offset (in this case from-beginning)
 - b. The broker streams back the messages. The consumer prints them in the command prompt.
 - c. Consumers track their progress using offsets.

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